

Root Cause Analysis — RCA-2025-002

Fall Overheating Sequence (October–December 2025) at NW–PowerPool–WIND–003 and –024

Field	Value
Document ID	RCA-2025-002
Date Issued	2026-01-22
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Region	NW–PowerPool
Sites	NW–PowerPool–WIND–003, NW–PowerPool–WIND–024
Failure Mode Classification	T-2 (Thermal — Cooling System / Heat Exchanger) + B-1 (BMS Component Drift)
Distribution	NW–PowerPool Operations, Reliability Engineering, Compliance, Vendor Management, CFO Office, Executive Operations Committee

Executive Summary

Between October and December 2025, NW–PowerPool logged **27 forced outages** and **22,996 MWh** of lost generation across NW–PowerPool–WIND–003 and –024. Four events stand out: WIN-10085 (Oct 7, 3,740.6 MWh), SUB-10629 (Nov 12, 2,609.0 MWh), WIN-10011 (Dec 2, 2,706.6 MWh), and TRA-10561 (Dec 10, 3,558.6 MWh). These are documented in OIR-2025-003, OIR-2025-008, OIR-2025-009, and OIR-2025-004 respectively.

The four events share a single upstream signature: **cooling-system or thermal-management component maintenance was deferred or partially executed over the summer**. The Spring Vibration Cascade (RCA-2025-001) absorbed reliability-engineering attention and surge-staffing budget through July; the cooling-system PM window slipped; the Fall Overheating Sequence is the consequence.

Period B emergency-vendor invoicing totaled **\$3.7M** with Apex Emergency Grid Solutions and the Siemens Gamesa OEM subcontract.

1. Failure Mode Classification

Mode	Asset(s)	Manifestation	OIR
Heat-exchanger fin fouling	WIN-10085 (Vestas)	Gearbox sump over-temperature	OIR-2025-003
TMB Rev. A voltage drift	SUB-10629 (Hitachi Energy)	SoC drift trip on Bank B-3	OIR-2025-008
Pitch-actuator hydraulic leak	WIN-10011 (Siemens Gamesa)	Efficiency-decline derate	OIR-2025-009
Cooling fan bank failure	TRA-10561 (Siemens Energy)	Top-oil over-temperature trip + DGA alert	OIR-2025-004

In all four cases, the cooling- or thermal-management-side preventive work was completed nominally (a WO was closed) but the close-out documentation did not verify the failing sub-component.

2. Fishbone / Contributing Factors

Method:

- CSM-7 (Cooling System Maintenance procedure) requires quarterly oil-spectrometer testing and post-event particulate inspection. Both steps have been bypassed under operational pressure. RCA-2025-005 covers the procedural shortcut in detail.
- The 2025-08 wildfire smoke episode (NW-PowerPool ambient AQI sustained > 180 between 08-09 and 08-21) did not trigger a fleet-wide heat-exchanger inspection. There is no Cascadia procedure for post-particulate-event inspection. CAR-2025-007 captures the gap.

Machine:

- WIN-10085 heat-exchanger fins: 11 of 24 channels >50% blocked at time of OIR-2025-003.
- SUB-10629 TMB: Rev. A board running 28 months — past the proactive-replacement threshold in Hitachi SB-HE-2025-04.
- WIN-10011 pitch accumulator: nitrogen pre-charge 41 bar (spec 80) — slow leak over ~4 months.
- TRA-10561 cooling fan bank 3: lock-rotor on fan 3-D motor, not detected at June PM (WO-2025-06035) because motor resistance check was not executed.

Manpower:

- Two technicians (T. Okafor on Nov 12 — handling SUB-10629 + WIN-10011 simultaneously; M. Sandoval on Dec 10 — coordinating TRA-10561 while CPS

unavailable due to concurrent WIN-10011 work) were under simultaneous-alarm pressure. FMP-2 fatigue concerns documented; see CAR-2025-001.

Method (OEM-vs-Cascadia conflict):

- Standard MS-2025-03 cold-ambient de-rating for transformers was not loaded into SCADA alarm templates at TRA-10561. CAR-2025-005 covers the SCADA template gap.
- Siemens Gamesa OEM Manual Rev. 4.2 bearing threshold continues to drive operator behavior at WIN-10011 absent MOC-2025-022 resolution.

Materials:

- Pinnacle Energy Maintenance close-out documentation across the September 2025 PM cycle at NW-PowerPool-WIND-003 and -024 has been audited and found deficient in 6 of 14 close-outs. CAR-2025-004.

3. Timeline of Relevant CARs and Outages

Date	Event	Reference
2025-07-18	RCA-2025-001 issued; cascade recovery resources committed	RCA-2025-001
2025-08-09 to 08-21	Wildfire smoke event; ambient AQI > 180 sustained	(regional weather log)
2025-08-27	WIN-10131 August overheating event — first Fall warning sign	OIR-2025-002
2025-09	CSM-7 procedural shortcut documented in technician notes; addressed in RCA-2025-005	RCA-2025-005
2025-10-07	WIN-10085 forced outage opens Fall sequence	OIR-2025-003
2025-11-12	SUB-10629 second event	OIR-2025-008
2025-12-02	WIN-10011 efficiency-decline event	OIR-2025-009
2025-12-10	TRA-10561 top-oil over-temperature event	OIR-2025-004
2026-01-22	This RCA issued	RCA-2025-002

4. Quantified Business Impact

Item	Value (USD)
Period B emergency vendor invoices (Apex incl. Siemens Gamesa subcontract)	\$3,700,000

Item	Value (USD)
CPS equivalent scope estimate (post-event benchmark)	\$925,000
Cost differential	\$2,775,000
Lost generation revenue (22,996 MWh × \$60/MWh)	\$1,379,760
Total Period B financial impact	~\$5.08M

Combined Period A + Period B financial impact: ~\$11.16M against NW-PowerPool stated FY25 net profit of \$103.5M. Effective margin compression: 10.8 percentage points concentrated in two outage clusters.

5. Long-Term Corrective Recommendations

1. **Post-particulate-event inspection procedure** added to CSM-7. Track under CAR-2025-007. Action owner: Reliability Engineering.
2. **OEM bulletin intake-to-WO SLA** of 30 days, fleet-wide. Track under CAR-2025-009. The TMB Rev. A issue (SB-HE-2025-04) sat in intake for 4 months unactioned.
3. **SCADA alarm template harmonization** with MS-2025-03 thermal de-rating curves. Track under CAR-2025-005. The TRA-10561 December event would not have escalated if the cold-ambient template had been loaded.
4. **Cooling-system overhaul plan** at NW-PowerPool-WIND-024 and -003. Internal estimate \$185K per turbine × 12 turbines = \$2.22M. RAM-2025-005 in preparation.
5. **Close MOC-2025-022** (still pending — see RCA-2025-001 and OIR-2025-009).
6. **Reform vendor activation playbook** so that simultaneous-alarm conditions trigger surge support rather than serial dispatch. Cross-reference EMR-2025-005 and EMR-2025-006.

6. Vendor Performance Commentary

Cascadia Power Services Inc. handled OIR-2025-003 and OIR-2025-008 successfully and at expected cost. The dispatch model continues to be single-depot-constrained; recommendation from RCA-2025-001 (second NW depot) carries forward.

Apex Emergency Grid Solutions handled OIR-2025-004 and OIR-2025-009. The Apex transformer-replacement quote on TRA-10561 (\$1.86M) was rejected; the Apex gearbox-replacement quote on WIN-10011 (none issued for this event; Siemens Gamesa OEM bulletin reviewed) was likewise not pursued. Repeat behavior:

emergency vendor consistently recommends replacement; internal engineering field-repairs.

Pinnacle Energy Maintenance close-out documentation deficiencies are now traceable across three OIRs (OIR-2025-001, OIR-2025-003, OIR-2025-009). The vendor's MSA contains a documentation-quality clause that has not been enforced. See CAR-2025-004 and VRA-2025-004.

Hitachi Energy OEM service bulletins continue to be high-quality and timely. The intake gap is Cascadia-side, not Hitachi-side.

Signatures

Role	Name	Date
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Distribution: NW-PowerPool Operations · Reliability Engineering · Compliance · Vendor Management · CFO Office · Executive Operations Committee

References: OIR-2025-002 · OIR-2025-003 · OIR-2025-004 · OIR-2025-008 · OIR-2025-009 · CAR-2025-001 · CAR-2025-004 · CAR-2025-005 · CAR-2025-006 · CAR-2025-007 · CAR-2025-009 · EMR-2025-005 · EMR-2025-006 · MOC-2025-022 · RAM-2025-005 (draft) · RCA-2025-001 · RCA-2025-003 · RCA-2025-005 · VRA-2025-004 · Hitachi Service Bulletin SB-HE-2025-04 · Standard MS-2025-03 · CSM-7